

12 Motion in a circle

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Circular motion changes velocity direction even at constant speed. The centripetal acceleration $v^2/r = \omega^2 r$ points toward the centre, and its resultant force is mv^2/r . Angular speed $\omega = 2\pi/T$. “Centripetal force” describes a resultant supplied by real forces, not an extra interaction.

Worked example: follow the reasoning

A 2.0 kg object moves at 3.0 m s^{-1} around a 1.5 m radius: acceleration = 6.0 m s^{-2} and resultant radial force = 12 N.

Quick check · multiple choice

A What direction is centripetal acceleration?

[1]

- A Tangential to motion
- B Away from centre
- C Vertical always
- D Towards centre

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 An object travels a horizontal circle of radius 0.80 m in 2.0 s. Determine angular speed and centripetal acceleration.

[3]

Final answer: _____

Answers and explanation

Quick check: D. Towards centre

Question 1 [3 marks]

$\omega = 2\pi/T = \pi \text{ rad s}^{-1}$ [1]; $a = \omega^2 r$ [1] = 7.90 m s^{-2} [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.